



DM - PYQ - 1 - DISCRETE MATHEMATICS JUNE/JULY QUESTION PAPER

CSE (Data Science) (Jawaharlal Nehru Technological University, Hyderabad)



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Note: This question paper contains two parts A and B.
i) **Part- A** for 10 marks, ii) **Part - B** for 50 marks.

- Part-A is a compulsory question which consists of ten sub-questions from all units carrying equal marks.
- Part-B consists of ten questions (numbered from 2 to 11) carrying 10 marks each. From each unit, there are two questions and the student should answer one of them. Hence, the student should answer five questions from Part-B.

PART- A**(10 Marks)**

- Determine whether the following proposition is TRUE or FALSE:
If $1+1=2$ or $1+1=3$, then $2+2=3$ and $2+2=4$. [1]
- Write the contrapositive of the statement "It rains whenever the home team wins." [1]
- Determine whether $A - (B - C) = (A - B) - C$ is TRUE or FALSE. [1]
- Give an example of a function, $f: \mathbb{Z} \rightarrow \mathbb{Z}$ that is 1-1 and not onto. [1]
- Draw a logic gate diagram for the Boolean function $F(x, y, z) = \overline{(x\bar{y})} + x\bar{z}$. [1]
- Is the set of integers under multiplication a semigroup? [1]
- How many permutations of all the seven letters A,B,C,D,E,F,G can be formed. [1]
- Find the coefficient of x^8 in the expansion of $(x^2 + 2)^{13}$. [1]
- Draw the digraph with adjacency matrix $\begin{pmatrix} 0 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 1 & 1 & 0 & 1 \\ 1 & 1 & 1 & 0 \end{pmatrix}$. [1]
- If T is a tree with 999 vertices, then T has how many edges? [1]

PART - B**(50 Marks)**

- Let $A(x)$ denote the statement "Computer x is under attack by an intruder." Suppose that of the computers on campus, only CS2 and MATH1 are currently under attack by intruders. What are truth values of $A(\text{CS1})$, $A(\text{CS2})$, and $A(\text{MATH1})$?
- Show that each conditional statement $[(p \rightarrow q) \wedge (q \rightarrow r)] \rightarrow (p \rightarrow r)$ is a Tautology by applying a chain of logical identities (Do not use truth tables). [3+7]
OR
- Using the rules of inference, show that the premises "All movies produced by John Sayles are wonderful" and "John Sayles produced a movie about coal miners" imply the conclusion "There is a wonderful movie about coal miners." [10]

- Let $f(n) = 2n + 1$. Is f a one-to-one function from the set of integers to the set of integers? Is f an onto function from the set of integers to the set of integers? Explain the reasons behind your answers.

- Let A , B , and C be sets. Prove or disprove that $A - (B \cap C) = (A - B) \cup (A - C)$. [5+5]

OR

5.a) Let $\mathbb{Z}$ be the set of integers and an integer $m > 1$. Define a relation R as aRb iff $m \mid (a - b)$. Show that this relation defines an equivalence relation on $\mathbb{Z}$.

b) Let $f, g, h: \mathbb{R} \rightarrow \mathbb{R}$ be defined as:

$$f(x) = 2x^3 + 5, g(x) = \cos(x) \text{ and } h(x) = x^3 - 1$$

Find $ho(gof)$ and $(hog)of$. Are they equal?

[5+5]

✗ Let A_m be a set of factors of a positive integer m , and the relation is divisibility on A_m . For $m = 30$, show that $(A_{30}, /)$ is a Lattice. Draw its Hasse Diagram. [10]

OR

7.a) Show that the arbitrary intersection of subgroups of a group is a subgroup of the group. Will the union of two subgroups also be a subgroup of the group?

b) Show that in any Boolean algebra, if $a \cdot x = 0$ and $a + x = 1$, then $x = \bar{a}$. [6+4]

✗ a) Suppose that we draw a card from a deck of 52 cards and replace it before the next draw. Using the indirect counting technique, determine in how many ways 10 cards can be drawn so that the tenth card is a repetition of a previous draw?

b) In how many ways can a hand of 5 cards be selected from a deck of 52 cards? [7+3]

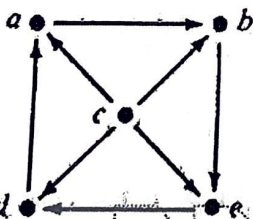
OR

9. Count the number of integral solutions of [10]

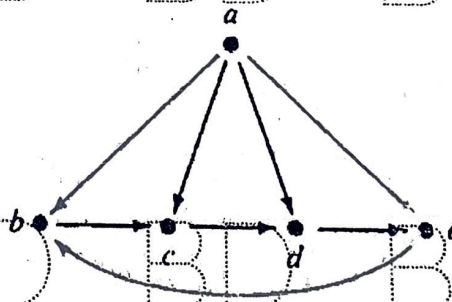
$$x_1 + x_2 + x_3 = 20 \text{ where } 2 \leq x_1 \leq 5$$

$$4 \leq x_2 \leq 7 \text{ and } -2 \leq x_3 \leq 9$$

✗ a) Define Isomorphic graphs. Determine whether the following two graphs are isomorphic or not. Justify your answer.

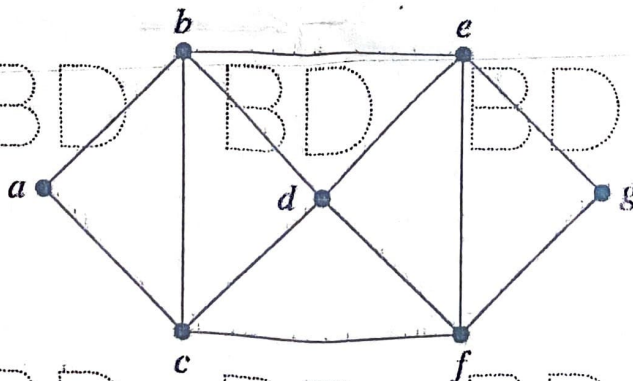


(a)



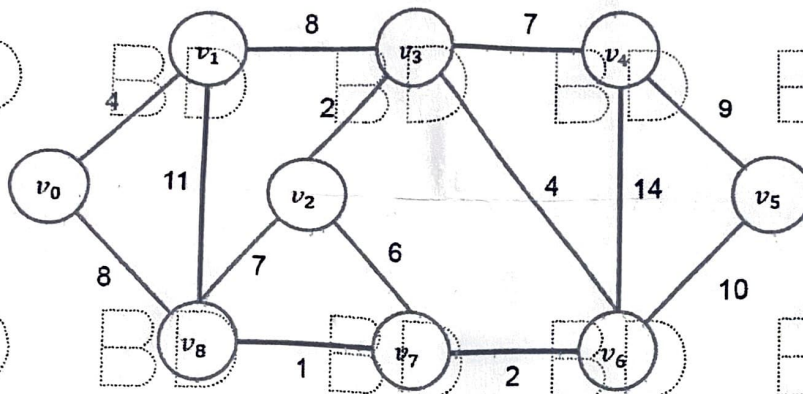
(b)

b) Determine the chromatic number of the graph G shown below. Also, draw the colored graph with vertices labelled with color names. [5+5]



G
OR

11. List the steps in finding the minimum spanning tree using Prim's algorithm. Using Prim's algorithm, find a minimum spanning tree for the graph shown below. [10]



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